FOSTER SCHOOL OF BUSINESS

UNIVERSITY of WASHINGTON

Paid Search Advertising Effectiveness at eBay

Background

Lakshmi Srinivasan, a senior marketing manager at eBay, stared at her dashboard of performance metrics. For years, her team had operated under a simple, largely unquestioned rule: allocate a fixed percentage of eBay's revenue to advertising. The logic was intuitive—spend money to make money. And in the digital age, much of that spend had naturally migrated toward online channels, especially paid search.

Paid search had always been the golden child of eBay's advertising portfolio. Unlike traditional media or even some digital formats like display ads, search advertising targeted users with clear intent. Someone typing "vintage Rolex" into Google was already in-market, likely just a few clicks away from making a purchase. If eBay's ad appeared at the top of that search page, a sale could be just seconds away. This visibility came at a price, but the prevailing wisdom was that it was worth it—after all, it reached high-intent users and claimed credit for the resulting transactions.

Over time, this belief solidified into budgeting practice. Paid search commanded a disproportionately large share of eBay's marketing budget, justified by its consistently strong performance on standard metrics like click-through rates, conversions, and return on ad spend (ROAS). The data seemed to speak for itself.

The Microsoft Incident

But the moment of doubt became unavoidable after a surprising turn of events. Earlier that year, tensions had flared between eBay's and Microsoft's marketing teams during a strategic partnership review. What started as a disagreement over campaign execution escalated into a broader rift, leading eBay to dramatically cut back its search advertising on Microsoft's Bing platform. As part of the fallout, eBay made the unusual decision to stop bidding on all *branded* keywords on Bing—terms like "eBay shoes" or "eBay guitar" that included eBay's name directly in the search query.

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The decision wasn't driven by performance data or careful analysis of ad spend. It was a reactionary cost-cutting measure, largely motivated by politics and frustration. Internally, the eBay marketing team braced for impact. Branded keywords had always been seen as a critical part of the paid search funnel—terms that signaled high purchase intent and generated reliable traffic and sales. By pulling those ads, the team expected to see a steep drop in traffic from Bing.

But that drop never came. To their astonishment, organic traffic surged to nearly fill the void. Web analytics showed that eBay retained about 99.5% of its branded keyword traffic on Bing even without running ads. Users who searched for "eBay guitar," still clicked—but now they clicked the unpaid organic links instead of the sponsored ones.

The implication was hard to ignore. Had eBay been paying to *cannibalize* its own traffic? Had years of marketing reports, dashboards, and ROI figures been overestimating the value of search ads simply because of how attribution worked? For Lakshmi, this was no longer a theoretical concern. The Bing episode had shaken her team's assumptions and raised a pressing question: If pulling back on paid branded ads didn't hurt traffic, what about the rest of their paid search portfolio?

Re-thinking Search Advertising Strategies

Lakshmi knew that a single datapoint—even one as striking as the Bing incident—wasn't enough to rewrite eBay's marketing playbook. But it had opened a crack in the foundation, and now she couldn't ignore the questions pouring through. If branded search ads weren't driving real incremental traffic, what about *generic* search ads?

These were the bread and butter of eBay's growth campaigns—ads shown when users searched for terms like "running shoes," "used laptop," or "vintage guitar." Unlike branded keywords, these queries didn't presume intent to visit eBay. In theory, that made them more valuable: they captured users still undecided, still browsing, still persuadable. But they were also vastly more expensive. Due to lower click-through rates and tougher competition, eBay often paid five to ten times more per click on generic terms than on branded ones. Lakshmi suspected that these ads were more likely to be incremental—but how much more? And more importantly, *were they worth it*?

The deeper she looked, the more she realized that her team lacked the tools to answer these questions with confidence. Their existing reporting relied heavily on last-click attribution—crediting the final touchpoint before a conversion, even if that touchpoint merely redirected users who were already intent on purchasing. That method systematically overstated the value of paid search, especially in cases where eBay would have captured the sale anyway via organic links, emails, or app usage.

What Lakshmi needed was a way to move beyond attribution—and toward causality. She didn't want to know how many clicks were associated with search ads. She wanted to know how many *sales would not have happened* without them. In other words, she needed to measure *lift*: the true incremental impact of paid search on eBay's bottom line. And this time, she wanted to measure it where it mattered most: *Google*.

Lakshmi faced two major challenges in moving forward. First, there was the risk. Google wasn't just another ad channel—it was *the* channel. It accounted for the overwhelming majority of eBay's

search advertising spend, and any disruption to that engine could have ripple effects across the entire business. Turning off or pulling back spend—especially without airtight guarantees—risked tanking traffic, revenue, and potentially even Lakshmi's credibility. The Bing episode had been an accident; taking action on Google would be deliberate. And if it went wrong, the consequences would fall squarely on her shoulders.

Second, there was the technical hurdle. While Google offered randomized controlled trial (RCT) capabilities for some of its ad products—such as YouTube, Display, and certain Demand Gen campaigns—those tools were not available for search advertising. And unlike other platforms, Google Search didn't allow advertisers to target or exclude users based on user IDs. There was no access to user-level data, and no mechanism for creating cleanly randomized test and control groups within the search environment. This left Lakshmi in a difficult position: she wanted to run a controlled experiment, but the platform she needed most didn't support it. There was no simple way to flip a switch and isolate the causal effect of paid search on user behavior. Any analytical approach would need to work around these constraints—without jeopardizing business performance or violating platform policies. Still, Lakshmi wasn't ready to give up. If she couldn't randomize at the user level, maybe there was another way.

Geo-Experiments

That's when Lakshmi remembered something from her school days—a method tucked away in one of her analytics lectures: *geo-experiments.* While most platforms, including Google Search, restricted access to user-level targeting or identifiers, nearly all of them still allowed advertisers to target based on geography. Search engines knew a user's approximate location—often down to the metro area—and used that information to deliver more relevant ads. Geo-targeting wasn't new in advertising. But using it to power a *randomized experiment at large scale*? That was a different story.

The principle behind geo-experiments was straightforward: if you couldn't assign individual users to treatment and control, you could assign *geographic regions* instead. In Lakshmi's case, that meant using eBay's 210 U.S. Designated Market Areas (DMAs) as the unit of analysis. Some DMAs could be randomly selected to turn off search ads, while others continued as usual. With this structure, she could observe the *causal* effect of removing paid search in treated regions—provided everything else remained stable.

The design wasn't perfect. In fact, it came with serious trade-offs. Most notably, the number of units was small. While online experiments typically involved millions of users, geo-experiments might only offer *hundreds* of units—each DMA acting as a single observation. This dramatically limited statistical power and increased the risk that random variation or local market shocks could obscure the true effect.

But Lakshmi also knew there were tools to mitigate that risk. Methods like *difference-in-differences* and *synthetic control* allowed researchers to extract more signal from fewer units by leveraging rich *pre-treatment data*. If the DMAs had stable trends before the intervention, Lakshmi could use that baseline to control for confounding variation and sharpen her estimates. These approaches wouldn't eliminate the challenges, but they could help her get closer to the truth—even in a world where user-level experimentation wasn't possible.

Planning

Over the next few weeks, Lakshmi set the wheels in motion. She partnered closely with eBay's Marketing Analytics team to hammer out the details of the experimental design. Together, they reviewed historical performance, seasonality, and market size across all 210 DMAs. The goal was to identify a balanced and representative set of "holdout" markets—DMAs where eBay would suspend all paid search advertising for the duration of the experiment. The rest would continue operating under the existing advertising strategy, serving as the control group.

Rather than pausing individual ads manually, the team architected the campaign structure in advance. They created distinct geo-targeted campaigns, isolating the treatment DMAs from the rest. That way, when the experiment launched, the holdout group could be cleanly removed from paid search simply by pausing the associated campaigns—no messy rewiring or manual overrides required.

But this wasn't just a data science project. Lakshmi knew that turning off search ads in dozens of markets—even temporarily—was a risky move. The decision could have real financial consequences. She brought in the Finance team early, working to define a set of *guardrails* that would allow leadership to monitor the experiment in real time. Together, they established thresholds for revenue impact, traffic loss, and campaign performance metrics that would trigger a pause or early termination of the experiment.

Lakshmi also engaged with executive leadership to secure alignment and buy-in. She wasn't asking for a long-term change—just a controlled test. Still, she needed everyone to understand what was at stake: the upside of uncovering millions in potential wasted spend, and the downside of temporarily sacrificing performance in key markets. Everything was nearly in place, and it was time to run the test.

Execution and Launch

After weeks of planning, Lakshmi and the team made a critical decision: the experiment would run during the second quarter of 2024. Q2 struck the right balance. It offered a more "typical" picture of consumer behavior compared to the extremes of other quarters. Running the test during Q4—the peak holiday season—was out of the question. Not only were customer behaviors highly seasonal, but the stakes were also too high. A failed experiment in Q4 wouldn't just be a missed opportunity; it could translate into massive revenue loss. Q1, on the other hand, tended to be sluggish—a hangover period after the holidays. There was concern that launching in Q1 might understate the value of search ads, simply because customers were buying less overall. Q2, by contrast, was stable—strong enough to matter, but not so critical that the company couldn't take a calculated risk.

The team structured the quarter into two phases:

- The **Pre-treatment period** would span the first 45 days. During this time, all DMAs would continue receiving their usual paid search ads. This baseline data was essential. It

would allow the team to compare trends across regions and account for any pre-existing differences between treatment and control DMAs.
- The **Post-treatment period**—the second half of the quarter—would serve as the heart of the experiment. At the start of this phase, Lakshmi's team paused all search advertising (both branded and generic) in the designated treatment DMAs. The control DMAs continued to receive their regular ad exposure.

By splitting the quarter in this way, the team could apply methods like difference-in-differences to isolate the causal effect of turning off paid search, using the Pre period to establish a solid comparison.

The launch went smoothly. Campaigns targeting the treatment DMAs were paused as scheduled. Real-time dashboards tracked traffic, revenue, and performance metrics across all regions. Importantly, none of the pre-defined guardrails were breached. There were no sudden collapses in revenue or traffic that would warrant halting the test.

For Lakshmi, it was a relief—and a milestone. eBay had successfully executed one of the largest geo-experiments in its history, collecting a rich dataset that could finally answer the question that had been haunting her since the Bing incident: *Were search ads really driving incremental sales?*

Data Description

Each *row* in the dataset represents a *single day of observations for one Designated Market Area (DMA*) during Q2 of 2024 (April 1 to June 30). In other words, the data is structured at the *DMA-day level*, making it suitable for panel data analysis. The goal is to observe how various market characteristics, ad inputs, and sales outputs evolve over time and across treatment and control groups.

Column Descriptions

- **DMA**: The name of the Designated Market Area (DMA), a geographic region used for advertising targeting and analysis.
- **Code**: A unique numeric identifier assigned to each DMA (e.g., 501 for New York).
- **Date**: The calendar date of the observation, covering the duration of the experiment (Q2 2024).
- **Population**: The estimated population of the DMA. This variable helps control for market size in the analysis.
- **MedianHHIncome**: Median household income in the DMA. Captures economic differences that may affect consumer spending behavior.
- **Broadband**: The percentage of households in the DMA with access to high-speed internet. Serves as a proxy for digital infrastructure and online shopping accessibility.
- **Urbanization**: A score representing the degree of urbanization in the DMA (e.g., higher values indicate more urban areas). Affects search behavior, device usage, and shopping patterns.
- **eCommIndex**: An internally developed index indicating the overall propensity of residents in the DMA to engage in e-commerce. Higher values reflect more digitally engaged markets.

- **MedianAge**: The median age of residents in the DMA. Useful for controlling demographic differences in online shopping preferences.
- **CollegeDeg**: The percentage of adults in the DMA holding at least a college degree. May correlate with digital literacy or brand awareness.
- **CostIndex**: A composite cost-of-living index for the DMA, scaled so that 100 represents the national average. Higher values indicate more expensive regions.
- **Treatment**: A binary indicator (0 or 1) denoting whether the DMA was assigned to the treatment group (ads turned off during Post period).
- **Post**: A binary indicator (0 or 1) for the experiment phase. 0 denotes the Pre-treatment period (all DMAs receive ads), and 1 denotes the Post-treatment period (when treatment DMAs stop receiving ads).
- **DayIndex**: A sequential day counter, starting from 0 on April 1, 2024. Useful for time series plotting and trend analysis.
- **AdSpend**: The dollar amount eBay spent on paid search advertising in the DMA on that day. For treatment DMAs during the Post period, this value is 0.
- **PaidClicks**: The number of clicks on eBay's paid search ads from users in the DMA on that day.
- **OrganicClicks**: The number of clicks from organic (unpaid) search results, capturing traffic that reached eBay without ad spend.
- **TotalClicks**: The total number of search-driven clicks, computed as the sum of PaidClicks and OrganicClicks.
- **SalesAds**: The dollar value of sales attributed to users who arrived via paid search ads. This helps in estimating the direct return from ad-driven traffic.
- **TotalSales**: The total dollar value of all sales in the DMA on that day, from both organic and paid traffic, as well as non-search sources.

Note: Due to technical glitches and data pipeline issues during the experiment, data from 10 DMAs had to be excluded from the final dataset. As a result, this dataset includes daily observations for 200 DMAs rather than the full set of 210 originally planned. While this slightly reduces the sample size, the remaining data still provides sufficient variation and power for meaningful causal analysis.

Questions

Question 1: Naive ROI Calculation (Pre-treatment Period Only)

Using only the pre-treatment period data (where Post = 0), calculate the naive return on investment (ROI) for eBay's search advertising. Make sure to aggregate over all DMAs in the pre-treatment period to calculate total AdSpend and total SalesAds. Then, in 2–3 sentences, explain why this naive approach may overstate the effectiveness of paid search advertising.

Question 2: Comparing Treatment and Control Groups

a) Using the pre-treatment period (where Post = 0), calculate and compare the *average daily TotalSales* for the Treatment and Control groups. What do you observe? What does this tell you about the comparability of the two groups before the intervention?

b) Now look at the post-treatment period (where Post = 1). Compare the average daily TotalSales between the Treatment and Control groups. Can we interpret the difference as the causal impact of search advertising? Why or why not?

c) Based on your answers to (a) and (b), summarize what conclusions (if any) you can draw about the effectiveness of paid search from a simple group comparison.

Question 3: Benchmarking Against Pre-Treatment Differences (DiD Intuition)

In the previous question, you observed that Control DMAs had higher total sales than Treatment DMAs in the pre-treatment period. This is likely due to structural differences such as market size, income levels, or online shopping propensity.

One way to adjust for these baseline differences is to use the *pre-treatment sales gap* between Control and Treatment as a benchmark. Then, examine how that gap changes in the post-treatment period:

- If *search ads have no effect*, we would expect the sales gap between Control and Treatment to stay the same.
- If search ads do have an effect, we would expect the gap to *widen* (Control does better) or *shrink* (Treatment catches up).

a) Using average daily TotalSales, calculate:

- The pre-treatment difference between Control and Treatment.
- The post-treatment difference between Control and Treatment.
- The change in this difference (i.e., DiD estimate of the causal impact of search ads).

b) What assumptions are we making in this approach? What could go wrong? List at least one potential threat to the validity of this estimation.

Question 4: Formal DiD Estimation Using OLS and Regression Enhancements

a) Run a simple OLS regression that replicates the analysis in Question 3.

b) Extend the regression from (a) by including relevant covariates that vary by DMA and could influence sales—such as Urbanization, MedianHHIncome, and E-Comm-Index. Choose variables you believe are important, justify your selection briefly, and compare the new DiD estimate to the result from (a). Did controlling for covariates improve the precision or change the estimated impact?

c) A common improvement in DiD analysis is to include *DMA fixed effects*, which account for any time-invariant differences between DMAs—like population, cost of living, or digital infrastructure—that don't change during the experiment. **Intuition:** There is substantial variation in DMA size and sales volume, even *within* the treatment and control groups. Some DMAs—like New York or Los Angeles—generate far more sales than smaller regions. When comparing the

overall treatment group to the overall control group, this creates a lot of noise and makes our estimates less precise, often leading to wide confidence intervals. By including DMA fixed effects (i.e., treating each DMA as its own intercept), we eliminate this noise by *comparing each DMA to itself over time*. This allows us to better isolate the impact of the ad suspension without being confounded by persistent regional differences in sales levels.

Re-run the regression from part (a), now including DMA as a categorical variable (i.e., fixed effects). Compare your estimated treatment effect to parts (a) and (b). How does it change, and what do you conclude?

d) Another improvement is to include day fixed effects (also known as time fixed effects). These control for common shocks or trends that affect *all DMAs on a given day*, such as national holidays, platform outages, or general weekday/weekend patterns. **Intuition:** Including day fixed effects helps strip out "background noise" and ensures that the estimated treatment effect isn't driven by shared time patterns across DMAs. For example, if overall sales dip on Sundays, that pattern won't distort the treatment effect if it's absorbed by day FEs.

Re-run the regression from part (a), now adding day fixed effects. How does the DiD estimate change compared to earlier models?

e) Finally, run the fully specified model with both DMA fixed effects and day fixed effects. This version gives you the most rigorous DiD estimate using the panel data.

Compare this final result to all previous parts (a–d). Does your estimate stabilize? Become more precise? Increase or decrease in magnitude? What does this suggest about the robustness of your findings?

f) In 3–5 sentences, summarize what you learned from this question. How did the inclusion of covariates and fixed effects change your interpretation of the impact of paid search ads? What did you learn about best practices in estimating causal effects using panel data?

Question 5: Estimating the ROI Using Causal Methods

In Question 4, you used a difference-in-differences (DiD) approach to estimate the **causal impact** of paid search ads on total sales. However, to calculate the true **return on investment (ROI)**, we also need an estimate of how much eBay would have spent on advertising in the treatment DMAs during the post-treatment period—*if ads had not been turned off*.

Just like sales, ad spending is likely influenced by seasonal patterns and DMA-specific characteristics such as population, income, broadband access, and education levels. Therefore, it is natural to apply the **DiD method again**, this time using **AdSpend** as the outcome variable, to estimate the counterfactual ad spending that would have occurred in the treatment group during the post-treatment period.

a) Use a DiD regression to estimate the amount of ad spending that would have occurred in the treatment DMAs during the post period, had the ads not been suspended.

b) Combine this counterfactual ad spending estimate with the incremental sales estimate from Question 4 to compute the **causal ROI** of the campaign:

c) Compare this causal ROI to the naïve ROI you calculated in Question 1. How different are they, and why? What do you conclude about the risks of relying on attribution-based ROI measures?

Question 6: Estimating the Impact Using the Synthetic Control Method

One of the more recent and advanced methods in causal inference is the **Synthetic Control Method (SCM)**—a sophisticated extension of difference-in-differences.

Intuition: The core idea of synthetic control is to create a *synthetic version of the treated unit* (or group) by combining multiple control units using optimized weights. These weights are chosen so that the synthetic control closely matches the treated unit's *pre-treatment trends* in the outcome variable (e.g., TotalSales), as well as important covariates (e.g., population, income, etc.). By doing so, it creates a credible counterfactual—i.e., what *would have happened* to the treated units if they had not received the treatment. In contrast to DiD, which assumes that all treated and control units would have followed the same trend absent treatment, SCM uses actual historical data to *construct* the most comparable control unit possible.

Using the Synthetic Control Method, estimate the causal impact of turning off search ads on total sales in Philadelphia DMA (Code 504).

Question 7: Conclusion

In a few well-developed paragraphs, summarize your key findings from eBay's paid search advertising experiment. What did your analysis reveal about the incremental impact of paid search on total sales? Was the return on investment (ROI) higher or lower than eBay might have expected?

Be sure to discuss any important *caveats* to your conclusion. For instance, consider potential biases, data limitations, assumptions behind each method, as well as potential implications in a business setting. Are the any important missing components not reflected in the analysis?

Finally, based on your findings, what would you recommend as the next steps for eBay's marketing and analytics teams? Should they cut back on paid search? Invest more selectively? What kind of follow-up analysis or testing would you propose before making major budget decisions?